

AI Supply Chain – BMC in AI Servers

June 2024

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Agenda

- Company Profile
- BMC is the Key to Optimize TCO
- BMC in AI Servers
- Product & Technology Development
- 2024 Q3 Outlook
- Q&A

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Company Profile

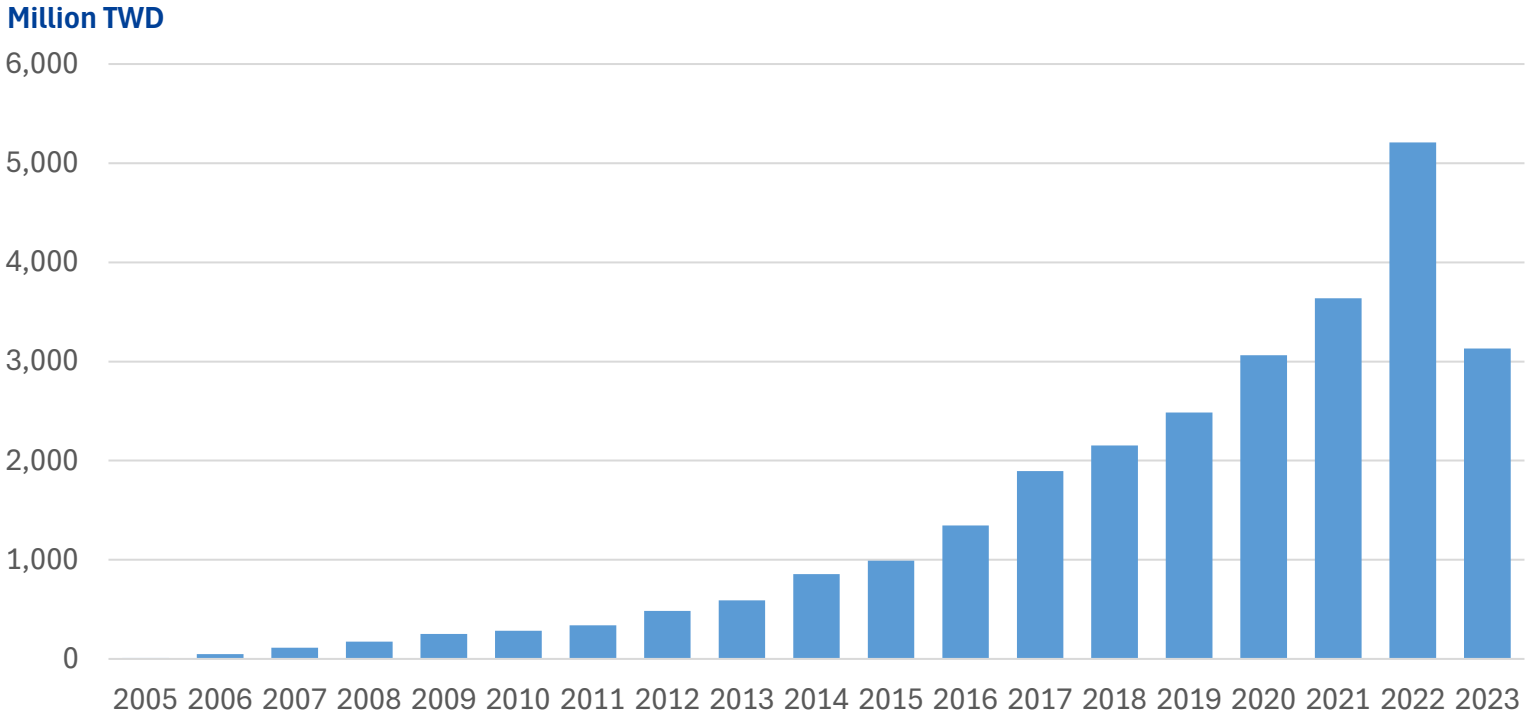
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Company Profile

- Founded in 2004
- Fabless-lite IC design company
- Number of employees: 125 (as of Nov. 2023)
- Product lineup (as of FY2023)
  - Cloud & Enterprise (BMC/BIC/PFR): 89% of revenue, **#1 in BMC**
  - Smart AV (Cupola360 & AVoIP SoC): 11% of revenue, **#1 in AV over IP (1G Ethernet)**
    #1 in industrial 360 camera
- Locations:
  - Headquarter: Hsinchu City, Taiwan
  - Branch office: Taipei City, Taiwan
  - US office: San Jose, CA

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Historical Revenue



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Milestones

2004.11

- **ASPEED founded**

2005-2006

- **Introduced 1st gen BMC SoC AST2000/1000**
- Delivered 1st mass production of SoC shipment

2007-2008

- ISO 9001:2000 certified
- Introduced 2nd gen BMC SoC AST2100/2050/1100
- Introduced 3rd gen BMC SoC AST2200/2150

2009-2011

- Launched new product line: PC/AV Extension SoC
- Received funding from Intel Capital

2012-2014

- Listed on the Taiwan OTC stock market
- **Listed on Taipei Exchange**
- Introduced 6th gen BMC SoC AST2500

2015-2016

- **BMC SoC market share ranked No.1**
- **Acquired Broadcom Emulex Pilot™ BMC SoC Business**
- ISO9001:2015 certified

2017-2018

- Joined Open Compute Project (OPC)
- Introduced new product line: Cupola360 Spherical Image Processor
- ISO 14001:2015 certified

2019-2023

- Introduced 7th gen 28nm BMC SoC AST2600
- ISO 27001:2015 certified
- ISO 14064:2018 certified

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Awards & Recognition

2014 - 2023

- **Forbes Asia's Best Under A Billion for 10 consecutive years**

2009

- COMPUTEX Best Choice Award

2008

- Ranked 37th in Deloitte Asia Pacific Technology Fast 500

2020

- Lenovo Technology Innovation Award
- Chairman & President Mr. Chris Lin received NYCU Distinguished Alumni Award



2021

- Chairman & President Mr. Chris Lin received Outstanding ICT Elite Award and NTHU Distinguished Alumni Award
- High-Growth Companies Asia Pacific 2021 from the Financial Times



2022

- COO Mr. CJ Hsieh received SEMI Industry Contribution Award and ISES Advisory Board Member Award
- Taiwan Best-in-Class 100 from Taiwan Institute of Directors and CDRC Consulting Group



2023

- Chairman & President Mr. Chris Lin received the ERSO Award
- InfoComm Best of Show
- Ranked 1st in Most Honored Company (All Asia Executive Team – Semiconductor) from Institutional Investor
- Ranked 5th in Business Performance of Taiwan's Top 5000 Large Enterprises from CRIF

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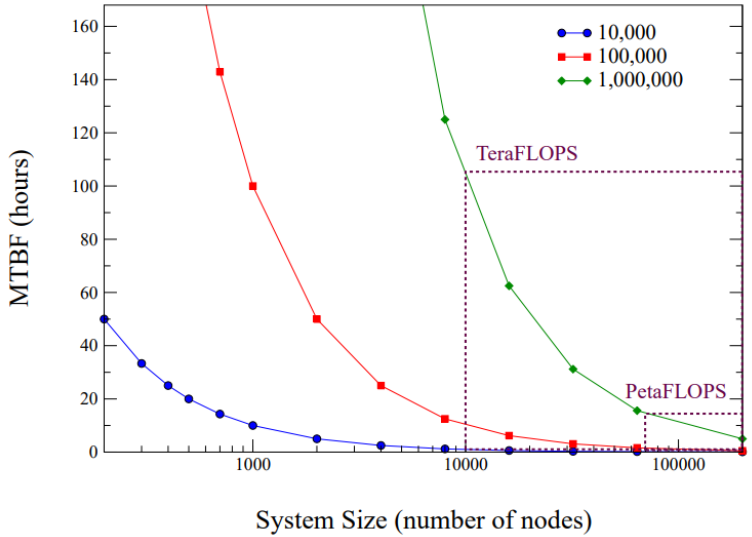
BMC is the Key to Optimize TCO

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Mean Time Between Failure

$$MTBF = \frac{1}{\sum_{i=1}^{i=N} \lambda_i}$$

- MTBF is the key to drive down Total Cost of Ownership
- λ_n : the failure rate of the component n

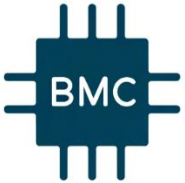


Note: MTBF stands for Mean Time Between Failure

Source: Fault-Tolerance in Large-Scale Parallel Machines with Buffered Co-scheduling, *Fabrizio Petrini, Kei Davis and Jose Carlos Sancho*

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BMC is the Key to Reduce TCO



- Error Detection and Logging
- Handling and Recovery
- Out-of-Band Management

RAS Engine

Optimal Total Cost of Ownership

AI-based preventative maintenance

- Maximize system uptime
- Improve resilience

Note: RAS stands for Reliability, Accessibility and Serviceability

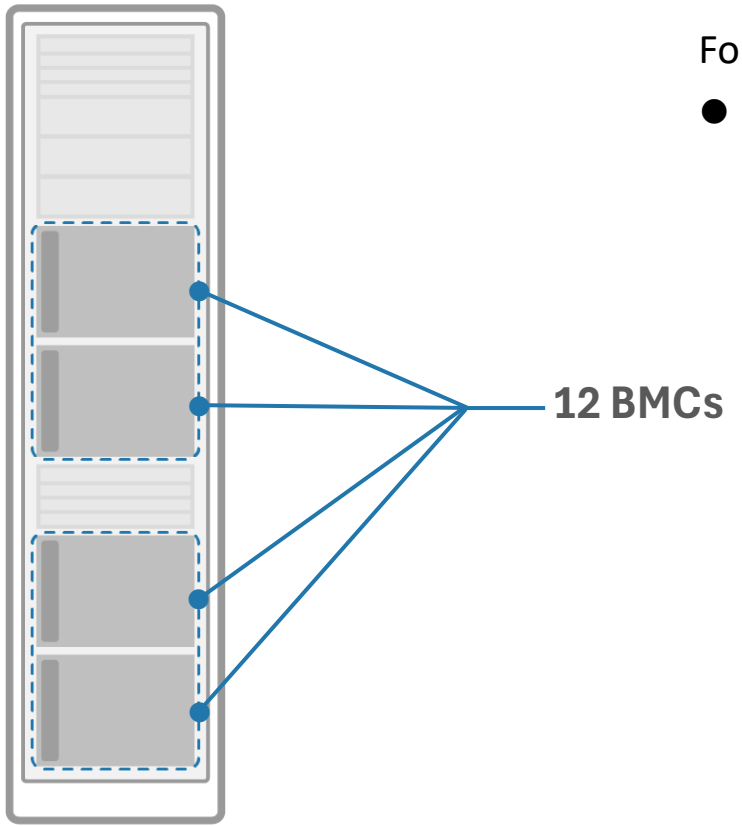
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BMC in AI Servers

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BMC as the Key Component in AI Servers

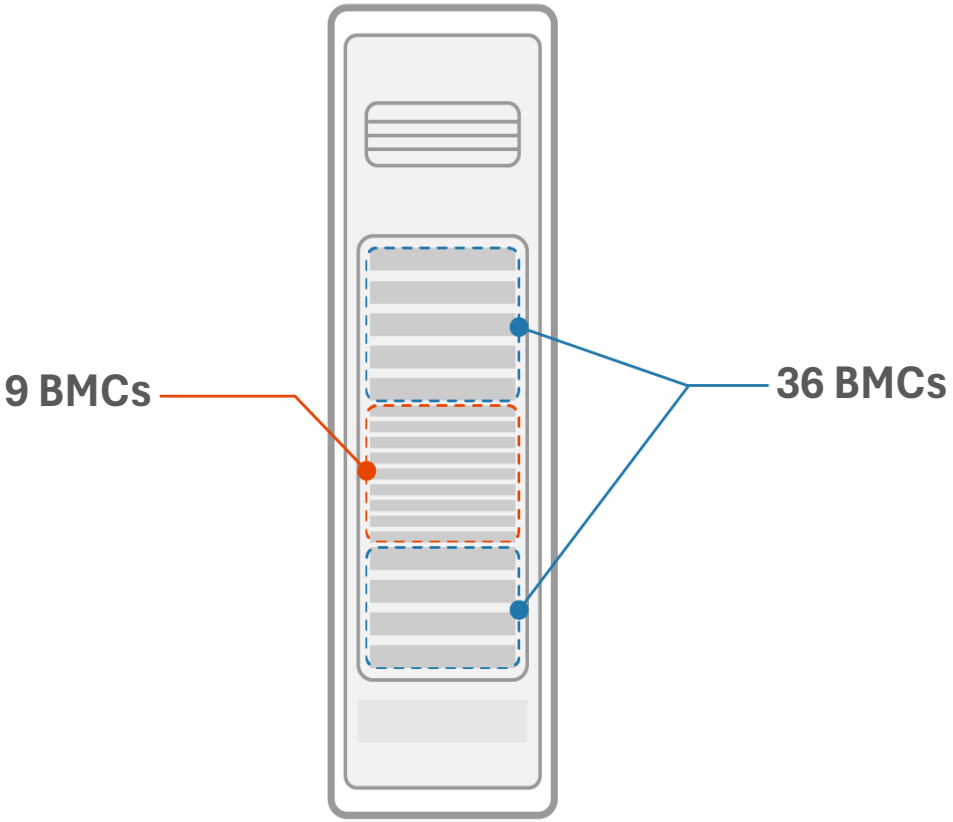


For Hopper HGX

- 12 BMCs in 4 compute trays

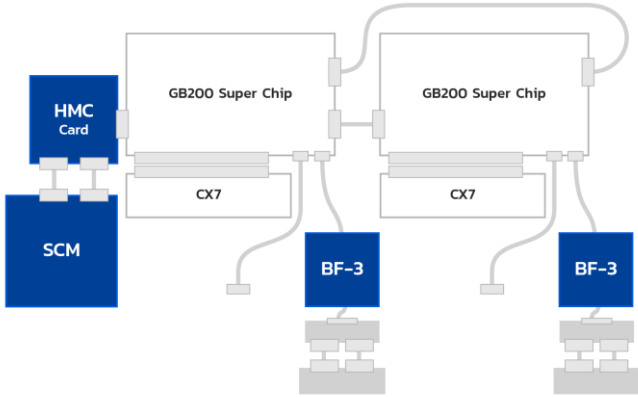
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BMC as the Key Component in AI Servers



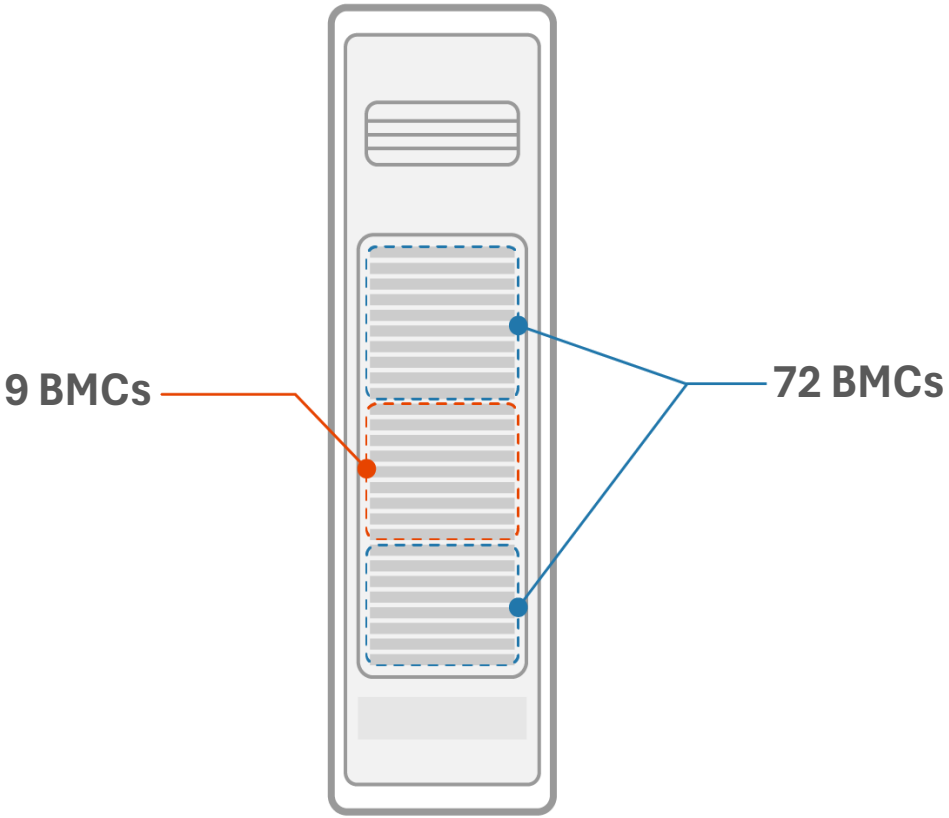
For Blackwell NVL36

- 36 BMCs in 9 compute trays
- 9 BMCs in 9 switch trays



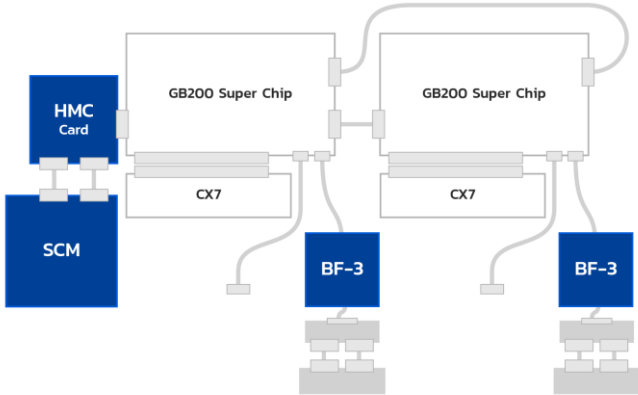
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BMC as the Key Component in AI Servers



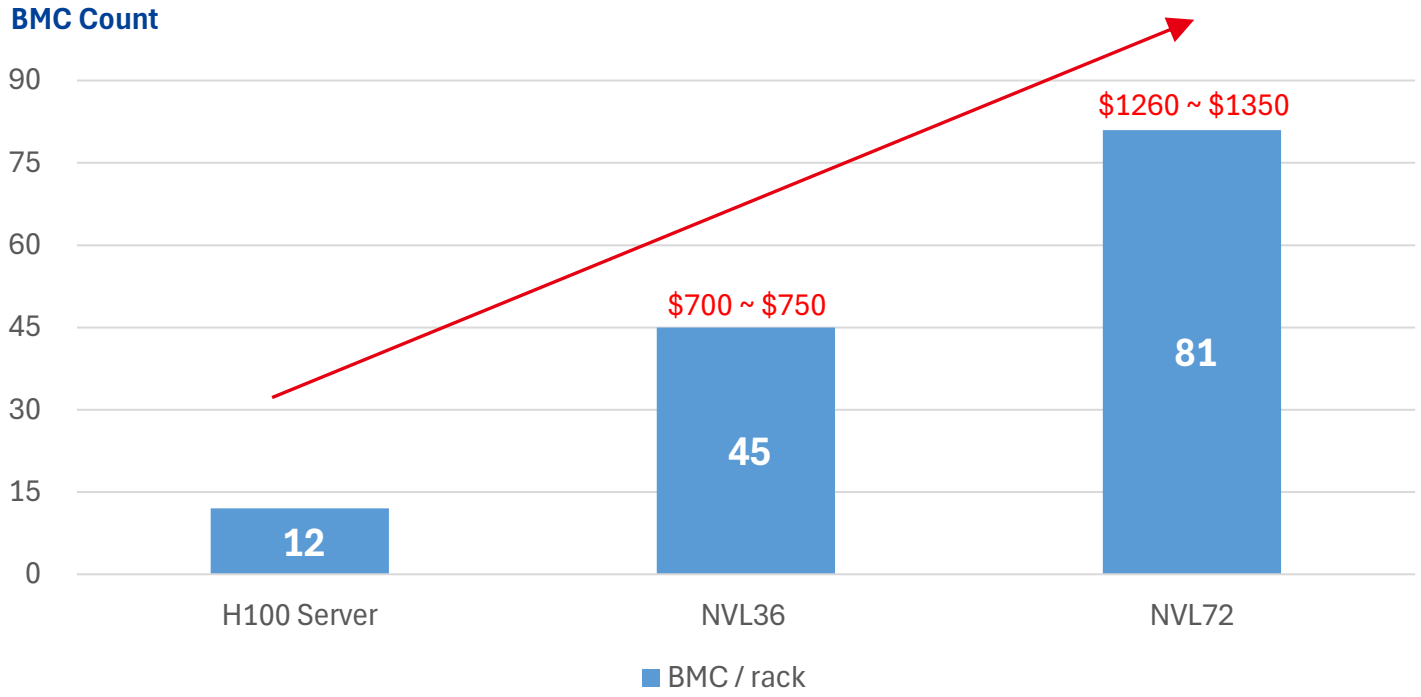
For Blackwell NVL72

- 72 BMCs in 18 compute trays
- 9 BMCs in 9 switch trays



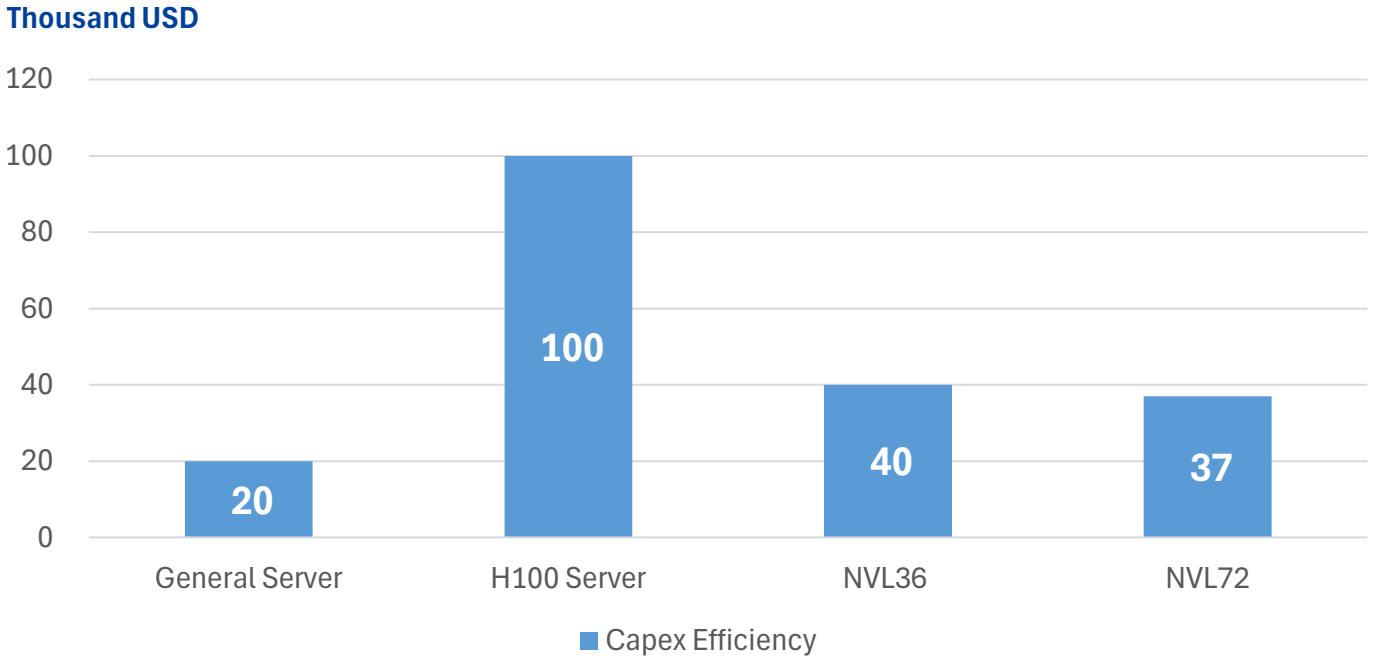
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BMC per Rack



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Capex Efficiency

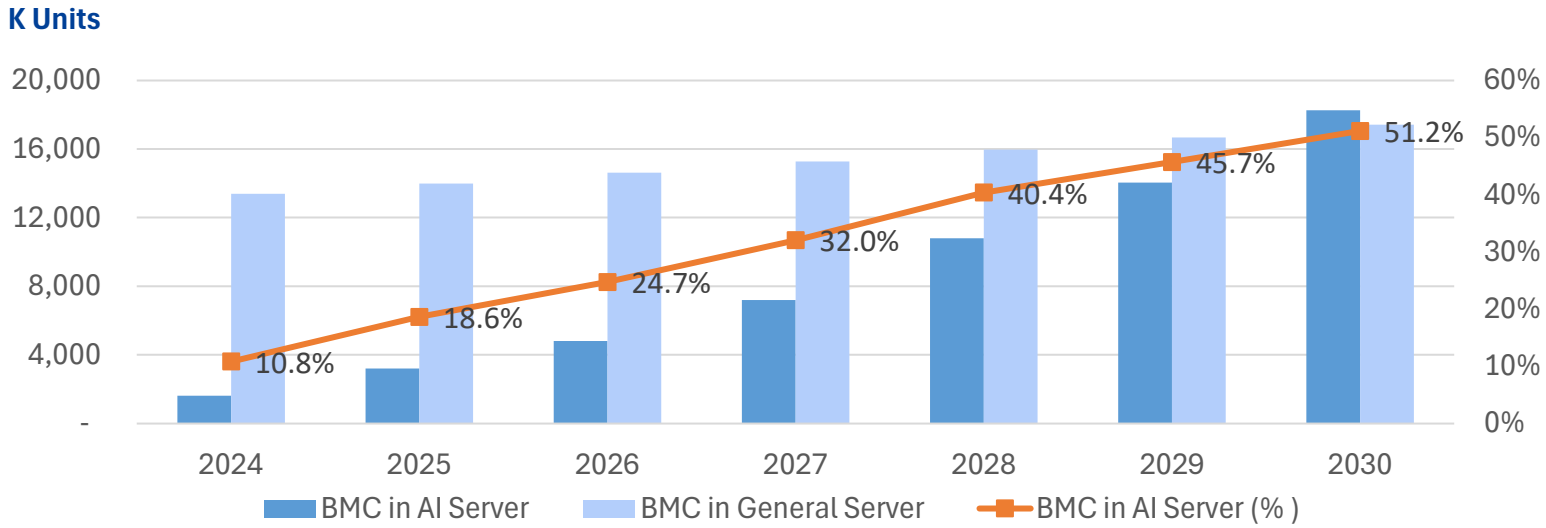


Note: Capex efficiency is defined as the amount of server spend that will contribute to one BMC demand

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BMC Market Forecast

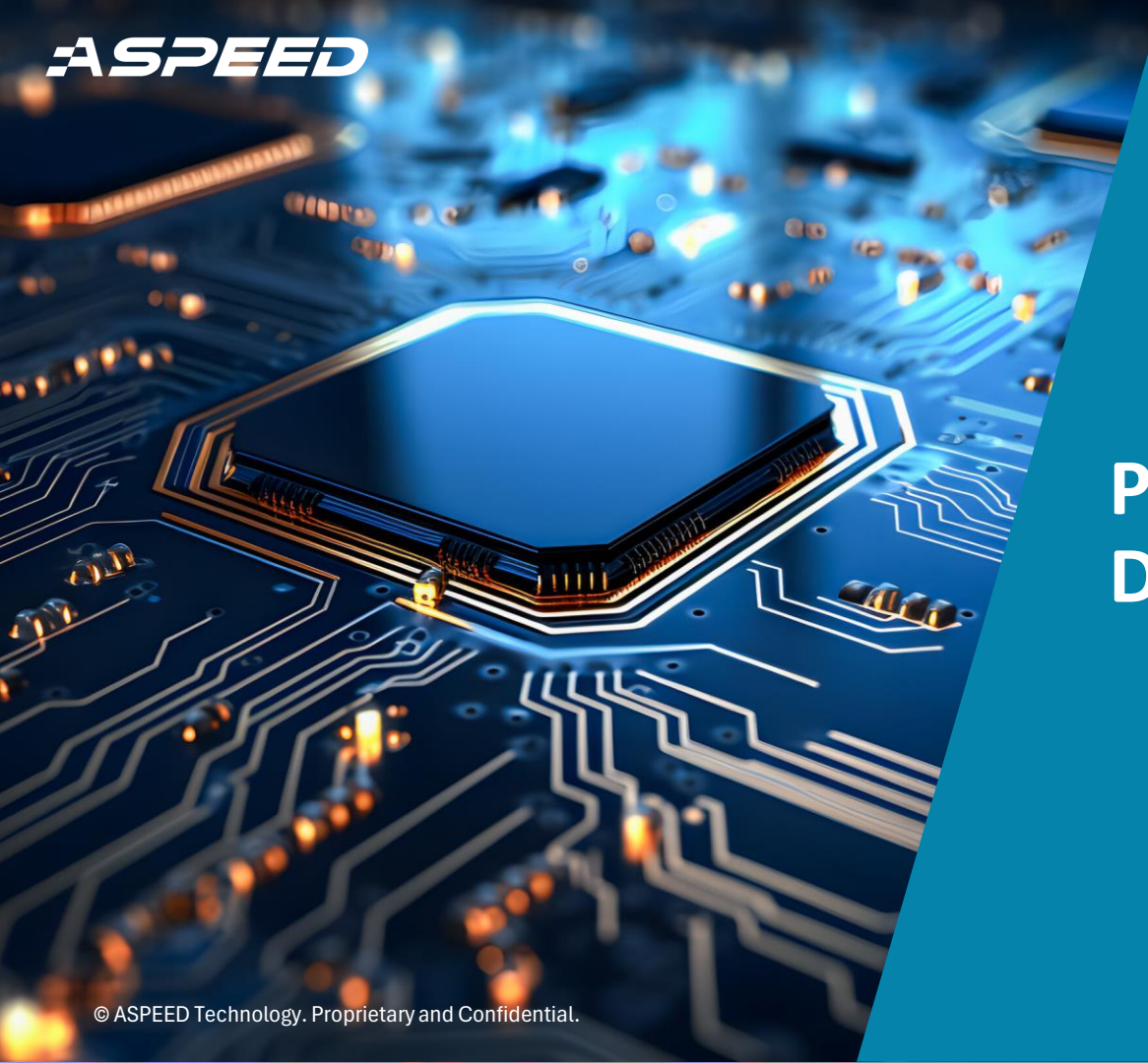
- BMC shipments for AI servers will surpass the BMC demand from general server in 2030.



Note: 1) BMC in AI Server (%) is the percentage of BMCs used in AI servers compared to the total BMC shipments.
2) BMC in general server will increase at a CAGR of 4.5%, while BMC in AI server will grow at a CAGR of 49.6%.

Source: ASPEED Technology estimate

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Product & Technology Development

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Product Portfolio



Cloud & Enterprise

- BMC：AST2500, AST2600, AST2700
- BIC：AST1030
- PFR：AST1060
- I/O Expander：AST1700

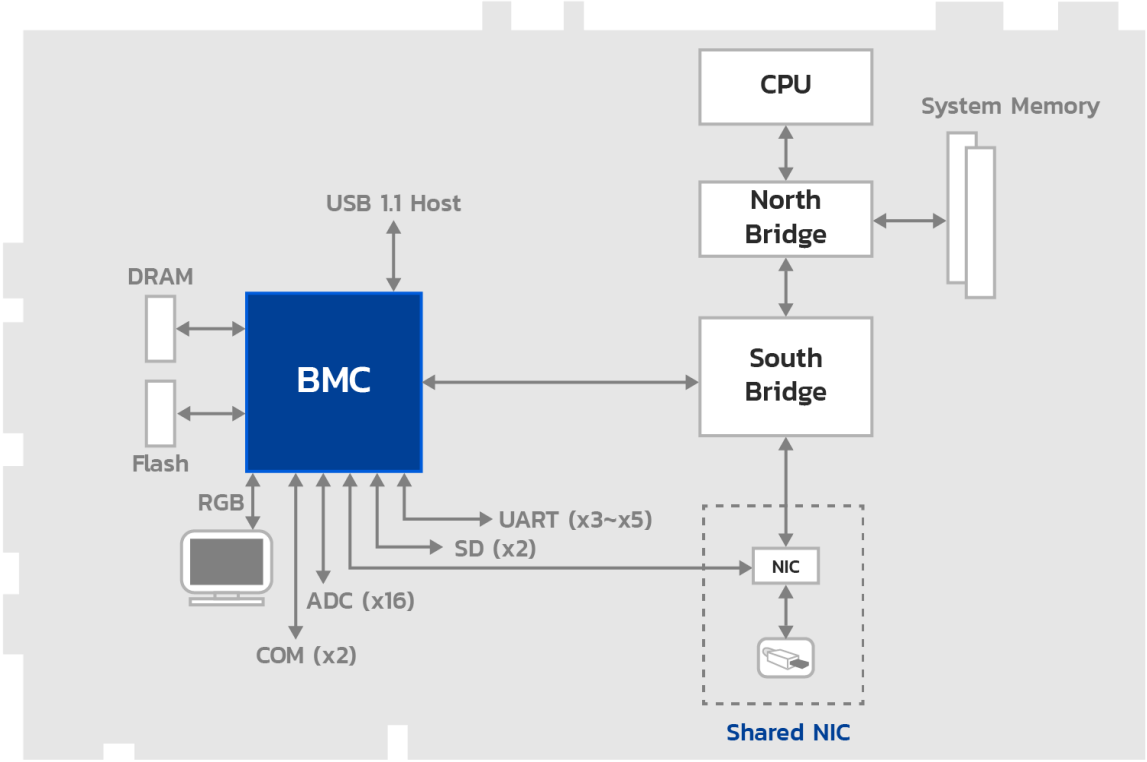
Smart AV

- AV Matrix：AST1520, AST1530, A1540
- AV Extender: AST1620, AST1630
- Cupola360：AST1220, AST1230, AST1240

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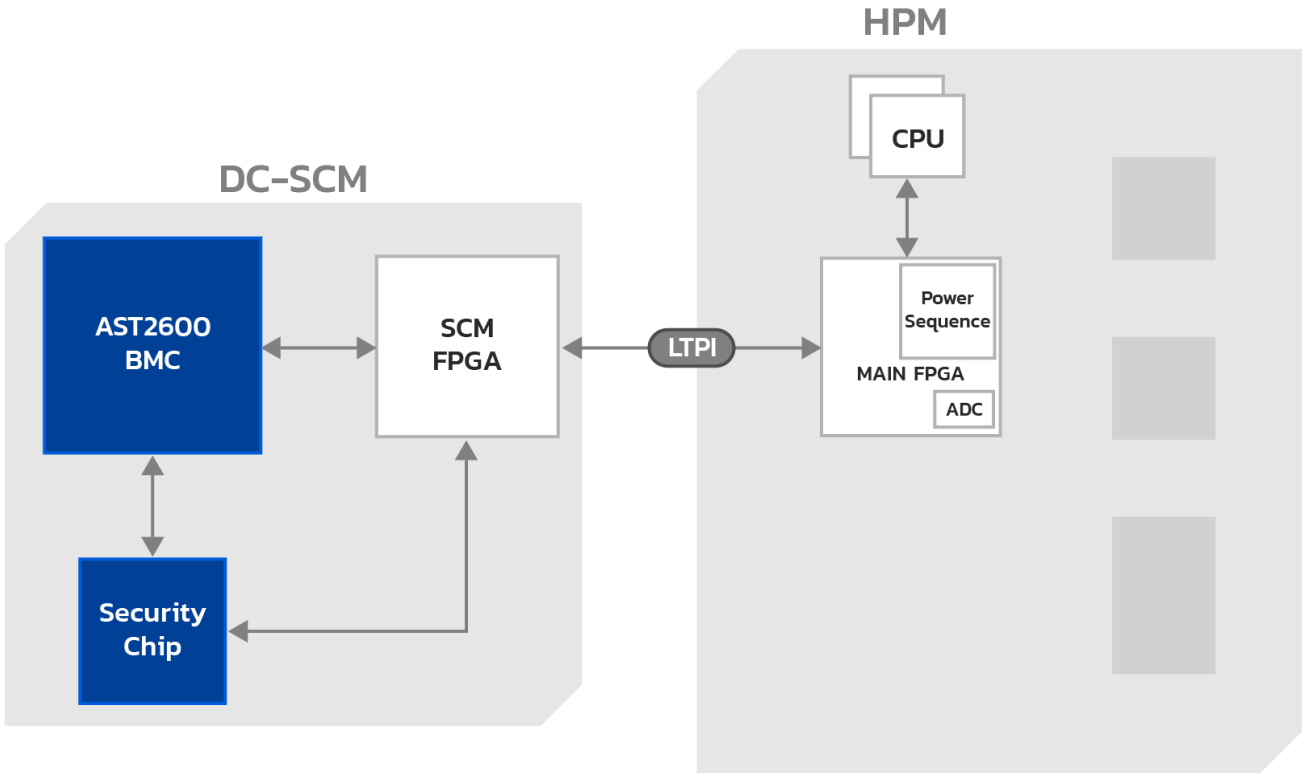


BMC in Traditional Server Architecture



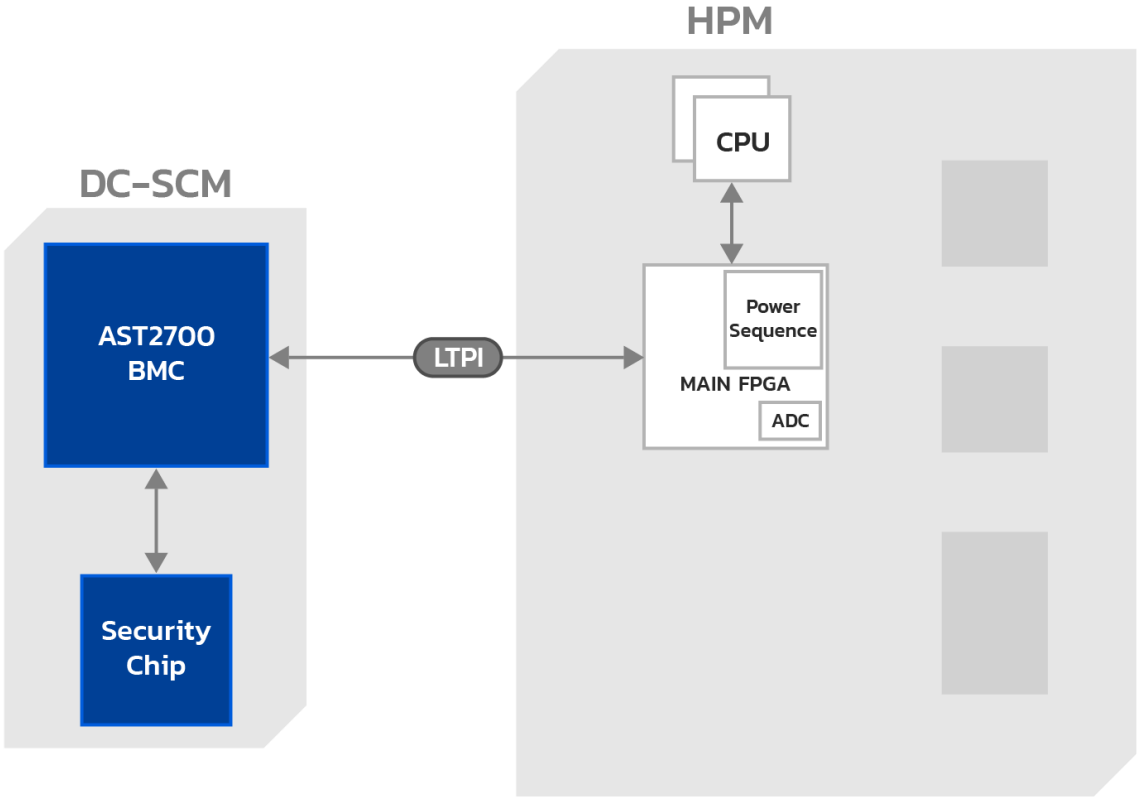
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Prevailing BMC in DC-SCM Architecture



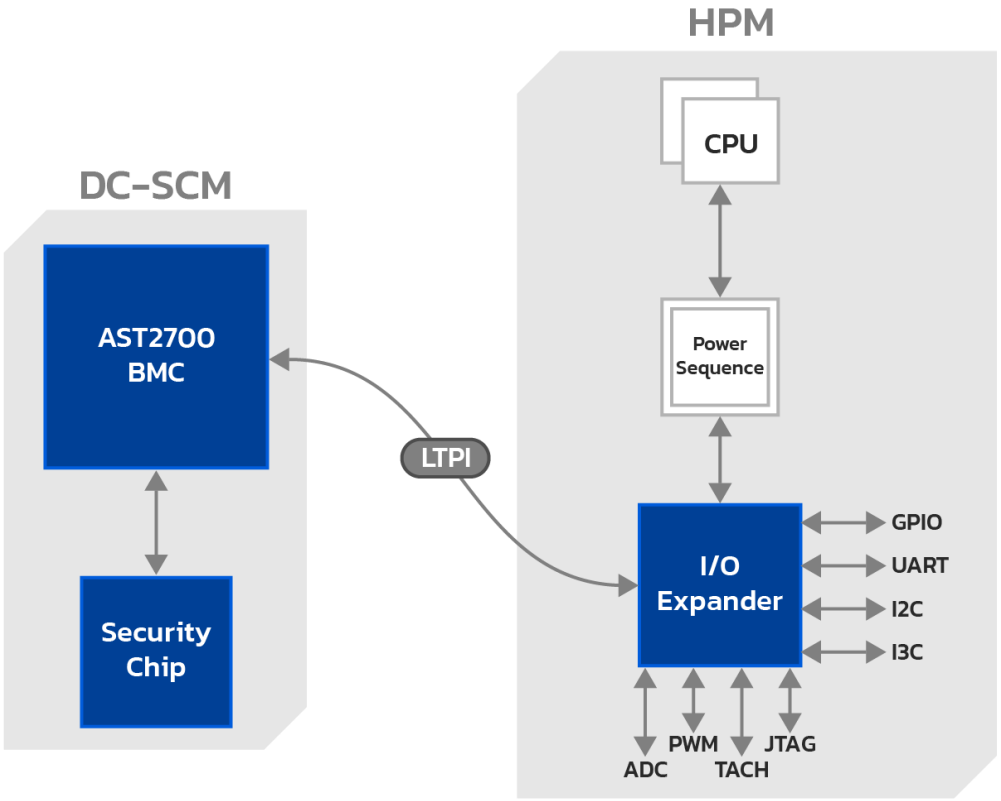
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Next Generation BMC



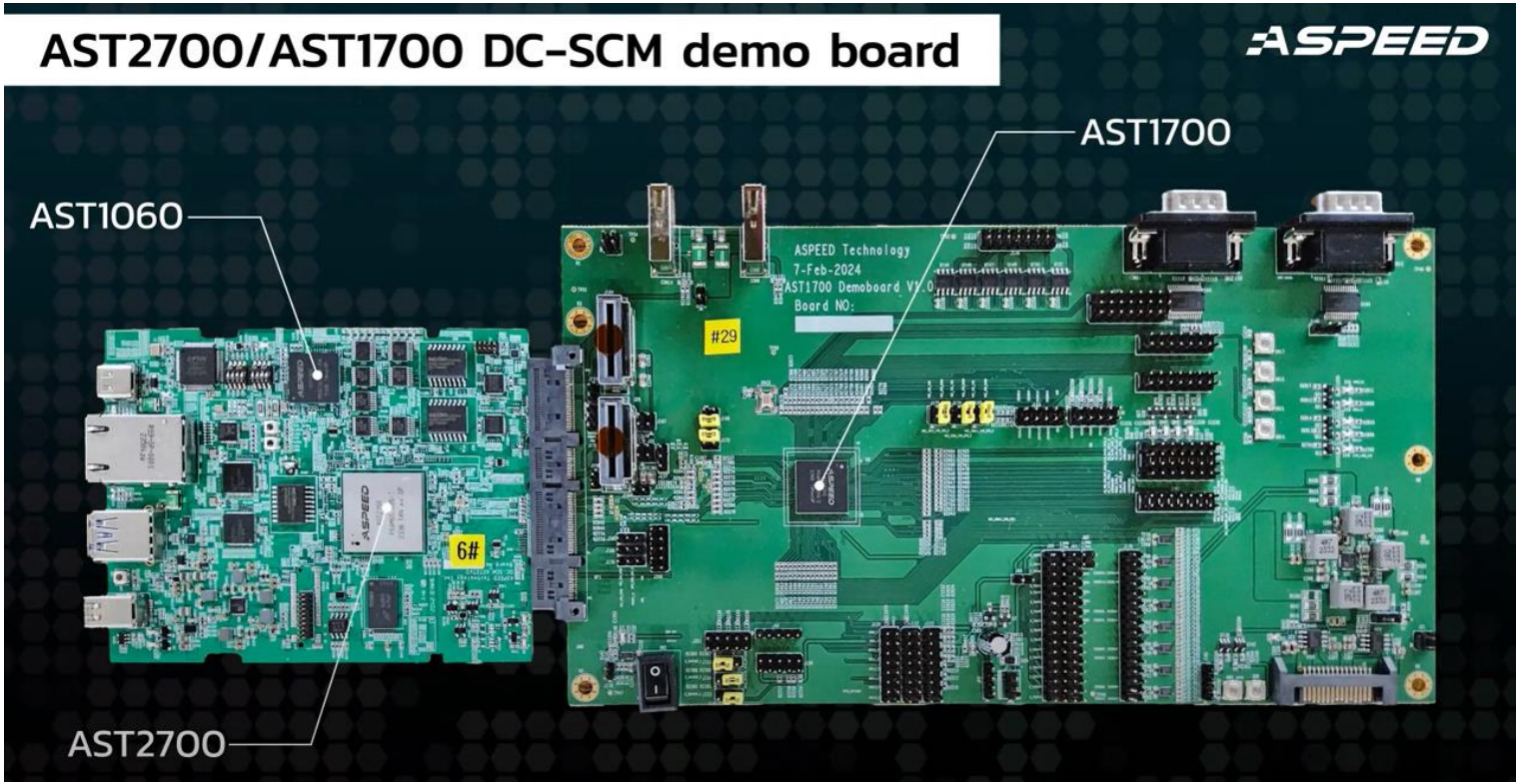
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Next Generation BMC with I/O Expander



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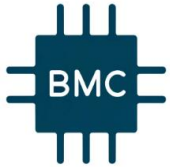
BMC with I/O Expander Demo Board



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AST2700 Enhancement and Innovation

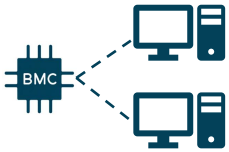




The most powerful BMC ever built (28x faster)

- 1 ARM Cortex® –A35 (Quda-core 1.6GHz per core)
- 2 ARM Cortex® –M4 (400MHz)
- 2 RISC-V CPU

∗ Multi-Thread benchmark between AST2600 and AST2700



The 1st BMC supports Dual-Node

- AST2750



Integration of Southbridge features

- USB 3.2 over PCIe xHCI



Compliant with OCP DC-SCM Standard

- LVDS Tunneling and Protocol Interface (LTPI)
- SiRoT (Caliptra)

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AST1700 Vantage Point



Simplify the H/W Design

- Firmware-less and LTPI interface connect to AST2700
- EIO (I3C, I2C, GPIO, JTAG, PWM, FAN Tech…)

Simplify the S/W Development

- No S/W development, upgrade, maintenance and protection efforts
- Seamlessly controlled by AST2700 series product

Differentiated Value-added features

- I3C Expansion and MCTP over I3C
- MCTP over SMBus (I2C)

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BMC Business Strategy

	General Server	AI Server
Trend	Low single-digit growth rate	Hyper growth rate
Strategy	Maximize dollar content per server	• Reduce Total Cost of Ownership (TCO)

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2024 Q3 Outlook

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Business Outlook

Cloud & Enterprise Solutions

- Capturing the opportunity of strong demand in AI servers.
- Mild recovery of general server shipments in 2024.
- Maximize the ***dollar value per server***
  - by higher BMC ASP and introducing Security Chip and I/O Expander

Smart AV

- AVoIP is expected to maintain its ***double-digit growth rate*** in 2024.
- Cupola360 is quickly deployed to smart factories and other use cases.

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Business Outlook

For 2024 Q3, we currently expect:

- Consolidated Revenue: Around **NT$1.6 billion ~ 1.7 billion**
- Gross margin: **63.2% ~ 64.7%**

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Q&A

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